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Radar interference into LTE base stations in the 3.5 GHz band

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ABSTRACT

We study the interference from a rotating shipborne radar system that spectrally and spatially coexists with a Long Term Evolution (LTE) cellular communications network in the 3.5 GHz band to investigate the feasibility of LTE deployment in the United States coastal metropolitan cities in that band. First, we simulate the radar systems with realistic operational parameters. Furthermore, we leverage a detailed 3GPP-compliant LTE simulation with a sophisticated air interface modeling and investigate sensitivity of LTE to radar interference in macro cell, outdoor small cell, and indoor small cell scenarios. We simulate the propagation conditions between the radar and LTE system by adopting the Free Space Path Loss and Irregular Terrain Model commonly leveraged by National Telecommunications and Information Administration (NTIA), to account for propagation, diffraction, and troposcatter losses that the radar pulses undergo before they reach the LTE system. As a performance metric, we evaluate the throughput of the LTE system in the uplink direction for various distances between the radar and the cellular system. Our simulation results indicate an LTE link will remain operational even in severe interference conditions. In fact, the LTE system as close as 100 km away from the radar undergoes less than 10% throughput loss from the LTE total throughput, and the throughput loss is less than 30% when the radar is only 50 km away from the LTE.

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1. Introduction

One additional tool available to Mobile Network Operators (MNOs) and other spectrum users in order to satisfy the mobile broadband communications data traffic increase [1] is spectrum sharing. Concerning this subject matter, the Presidents Council of Advisers on Science and Technology (PCAST) suggested leveraging government-held spectrum for mobile broadband needs. PCAST's proposal spurred the issuance of a Notice of Proposed Rulemaking (NPRM) [2] followed by rules by the Federal Communications Commission (FCC) proffering

the 3550–3700 MHz, abbreviated as the 3.5 GHz band, for mobile broadband purposes. The 3.5 GHz band is used by Department of Defense (DoD) radar systems which are entitled to full protection for their operations within their deployed areas. Soon after, the NTIA led a measurement campaign indicating the under-utilization of the 3.5 GHz band by the incumbents; this study revealed the untapped potential of the aforementioned band to be shared with non-federal communications.

A successful spectrum sharing precludes any destructive interference amongst the incumbent and entrant. Concerning this, NTIA performed a link budget analysis to investigate the interference between radar and commercial systems [3] in the 3.5 GHz band and provided with exclusion zones in its 2010 Fast Track Report [4], extending until 557 km inland and covering 60% of the United States (US) population as depicted by the yellow line in Fig. 1 based on technical assumptions available at the time.

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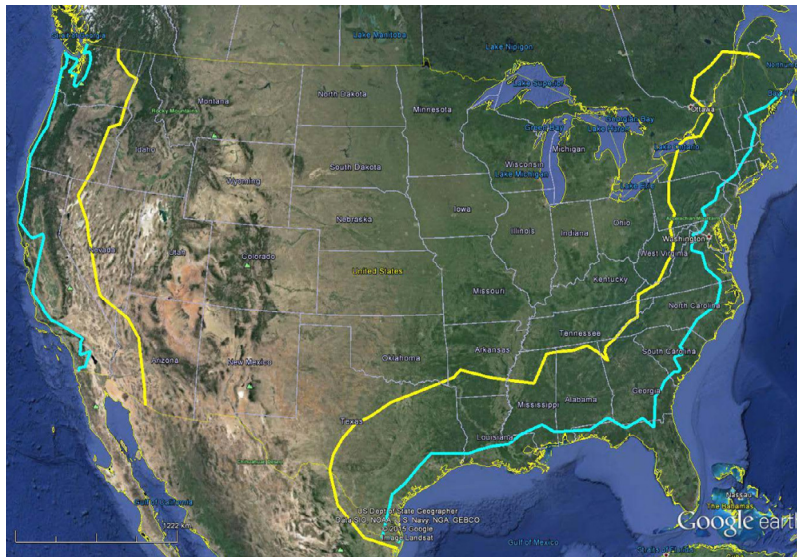


Fig. 1. NTIA simulations have proposed exclusion zones, Yellow Line—2010 Fast Track Exclusion Zones [4] and Blue Line—2015 Revised Exclusion Zones [5]. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

On March 24, 2015, NTIA filed a letter with the US spectrum regulator, FCC, recommending a framework that would reduce the geographic area of the zones by approximately 77% as depicted by the blue line in Fig. 1, based on new analysis of potential interference from commercial systems into incumbent radar systems which need to be protected. However, the reduced exclusion zones still translate into 40% of the US population being impacted.

Moreover, in its Report and Order [6] released on April 21, 2015, the FCC rightfully noted that NTIA's latest analysis effort to reduce the exclusion zones did not consider the potential interference impact to commercial systems from federal radar systems and encouraged commercial device and infrastructure manufacturers to study the issue and design equipment that overcomes or avoids harmful interference from federal radar systems. This paper presents such an analysis of interference from radar into commercial systems.

Because LTE is expected to be the prevalent cellular technology in the 3.5 GHz band, we studied the effect of interference from Federal radar systems into LTE. We concentrated on the UL direction because uplink interference could disturb the base station (BS) receiver, leading to degradation of service of the sites entire service area while downlink interference which is interference to the User Equipment receiver could have more limited impact on the communication quality of the whole system. Also, we concentrated on the performance of Time Division Duplex (TDD) LTE Systems that are expected to be deployed in this band.

In the course of our investigation, we simulated macro cell, outdoor small cell, and indoor small cell LTE deployments at the system level and compliant to the 3rd Generation Partnership Project (3GPP) [7]. Furthermore, the radar system simulation includes rotation, antenna pattern, dwell time, and other technical parameters for radar data sheets. Moreover, free space path loss (FSPL) [8], and irregular terrain model (ITM) diffraction

and troposcatter losses [9] are simulated using the parameters from the NTIA [4]. Our simulation results show that under the assumptions made in our study, LTE is very robust to radar interference even when LTE is experiencing extremely low signal-to-interference-plus-noise ratio (SINR) conditions. The current generation of military ship-borne radar operating in 3550–3650 MHz is used on only 17 ships deployed around the world. About 75% of this total are home-ported in Norfolk, VA, San Diego, CA, and Seattle, WA. Even if on exceptional occasions, multiple ship-borne radars are deployed near a given coastal area, it is unlikely that they will be using the same channels. Therefore, the probability of multiple radars around a given coastal location using the same channels as or adjacent channels to the LTE system is low. As such, it is reasonable to assume that the most likely scenario is that of one ship-borne radar interfering with the LTE system.

In the next section, we present the state-of-the art literature about the spectrum sharing between radar and mobile broadband systems.

2. Related work

In [4], NTIA investigated the interference between WiMAX and radar systems in the 3.5 GHz band by means of link budget analyses done on ecosystems containing radars on one side and WiMAX on the other side. The authors concluded that large geographic separations, reaching 577 km inland, between the two systems are required to allow for their operation. In [5], NTIA refined their analysis which led to a reduction of 77% of the exclusion zones but still hindered extending 3.5 GHz mobile broadband to the metropolitan areas in the US coastal areas. However, only a precise simulation of the involved technologies via simulating their protocol details can yield in relevant performance impact metrics due to interference between the systems. Next, Cotton et al. [10]

performed a measurement campaign where a shipborne radar in San Diego littoral waters was operating in the 3.5 GHz band. They measured the temporal mean channel occupancy over a two week period and found out that the 3.5 GHz band was not often occupied by the radar transmissions, thereby, underlining the staunch potential of the band for spectrum sharing applications.

Then, Sanders et al. [11] set up an experiment with Radio Frequency (RF) cabling to observe the interference from various radar waveforms produced by a signal generator and interjected into a 3.5 GHz TDD LTE evolved Nod-B (eNB). They looked at the throughput loss and block error rate (BLER) of the LTE system and realized that some waveforms did not show any appreciable effect on the LTE performance, whilst others undermined the LTE performance drastically. Next, Juskauskas et al. [12] studied the interference from radars to LTE systems in the 1.8 GHz band in the UL and downlink (DL) directions; they also showed that the radar interference lowers the LTE throughput. The authors in [13] evaluated the effects from a radar interferer into Voice over LTE (VoLTE) service of LTE systems in the 800 MHz band. The authors did not include any propagation model or detailed radar simulation in their studies.

The authors in [14] suggested a general spectrum sharing scheme based on time, space, frequency, and system-level modifications. However, due to significant amount of system modifications that they were proposing, their method was unconvolutive to real-world implementation. Then, the authors of [15] proposed projecting a radar signal onto the null space of the interference channel to allow for spectrum sharing with futuristic radars and conventional communications systems. However, their proposed method was not workable in the realm of the conventional radars currently operating. Awais et al. [16] performed a mathematical analysis of the communication transmitters (TXs) interference into a radar receiver (RX). But, they did not consider any radar antenna pattern or relevant propagation loss models. Ultimately, Ghorbanzadeh et al. [17] introduced a convex optimization approach to spectrum sharing between radar systems and sectorized communications systems with frequency reuse.

However, neither did they consider any propagation loss models, nor did they simulate a full-fledged radar system, nor their communications system was consistent with LTE frequency use patterns. In [6], the authors studied the impact of LTE small cell deployments on radar systems and showed that the exclusion zone can be less than 10 miles inland. However, they did not study the impact of interference from radar systems to LTE-based small cells.

2.1. Our contributions

Our contributions in the current article is as follows:

- We simulate a radar system with detailed technical operating parameters obtained from radar design equations for ship-borne S-band radars.
- We simulate the ITM propagation channel models respectively relevant to the LoS and NLoS regions according to which the radar signals are attenuated properly before reaching a cellular system.

- We simulate a full-fledged 3GPP compliant LTE system operating in the 3.5 GHz band and under a 10 MHz FDD and 20 MHz TDD mode.
- Our LTE simulation includes macro cells, outdoor small cells, and indoor small cells which are each simulated in accordance with the 3GPP documentations.
- We simulate the interference from the radar into the LTE systems in the UL direction by looking at the throughput loss the LTE system undergoes under the radar emissions.
- The interference results for both cochannel and out-of-band radars are investigated.
- Our simulation results show that LTE is very robust to radar interference even when LTE is experiencing extremely low signal-to-interference-plus-noise ratio conditions and indoor small cells are the most robust.

2.2. Organization

The rest of this paper proceeds as follows: Section 3 presents the background information for LTE, radar, and propagation models, needed to understand the current article presents the simulation for radar system. Section 4 presents the simulation for the outdoor macro, outdoor small cell, and indoor small cell LTE and radar systems. Section 5 presents the simulation results and analysis thereof. And, Section 6 concludes the paper.

3. Background

In the following subsections, we present the foundations of radar systems and their operations as well as those of the LTE systems necessary to understand this paper.

3.1. Radar preliminaries

Monostatic pulsed radar systems, referred to simply as radars in the scope of the article at hand, subsume high power radio TX and sensitive radio RX systems located at the same physical location and isolated via a transmit/receive (T/R) switch [18] connected to an antenna. A simplified block diagram for a radar system is depicted in Fig. 2. The radar antenna rotates in azimuth and elevation and the amount of rotation depends on the search fence which is specified by the tactical operation requirements [18]. Typical radar operating parameters are illustrated in Table 1. Conventionally, the azimuth rotation is 360°. While the antenna rotates, pulses are emitted out of the antenna into the space. The radar antenna pattern specifies the effective isotropically radiated power (EIRP) from the radar antenna. At the boresight direction, the maximum gain is added to the radar pulse powers; at other angles off the boresight, less gain will be compounded to form the EIRP. If the radar signals hit an object in the outer space, the object acts an omnidirectional electromagnetic source and reflects pulses in all directions in the space [18]. The echo pulses which are received by the radar receiver are decoded, the small pulse in Fig. 2, which reveals that a target is detected.

Radar peak power refers to the maximum pulse power which is the multiplication of the pulse amplitude

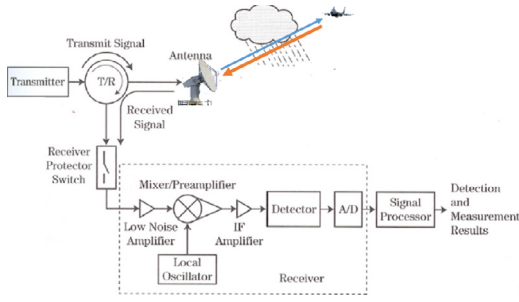


Fig. 2. Radar system subsumes collocated TX and RX separated with a T/R switch. The antenna rotates and transmit pulses into outer space, which are attenuated by propagation effects. If the pulses hit an object, the echos return to the radar and are detected by the radar.

by the pulse width, and the average power refers to the multiplication of the pulse amplitude by the pulse repetition interval (PRI), which is the radar pulse period. The actual power radiated out of the antenna is the addition of the radar peak power and the antenna gain in the direction under which the radiation power is of interest. Typical 50 kW–2 MW peak pulse power, i.e. 77–93 dBm, can lead to 1.6–2 GW, i.e. 122–138 dBm, EIRP. The antenna azimuth and elevation beamwidths θ_a and θ_e as well as the radar antenna horizontal and vertical scan angle θ_H and θ_V yield in M beam positions to form the search fence as Eq. (1). The ceiling function captures the fact that the number of horizontal and the number of vertical scan positions are positive integers.

$$M = \left\lceil \frac{\theta_H}{\theta_a} \right\rceil \times \left\lceil \frac{\theta_V}{\theta_e} \right\rceil. \quad (1)$$

The antenna rotation speed S , the number of rotations in a minute, generates an antenna dwell time T_d as in Eq. (2). The dwell time T_d is the amount of time that radar antenna spends in each beam position to be able to cover the total M beam positions forming the search fence at the antenna speed S . The typical values for these parameters are in Table 1. Pulse periods, aka the pulse repetition interval (PRI), T_p in radar terminology characterizes the number of pulses n_p are transmitted during the antenna dwell time T_d in accordance with Eq. (3). The relationship between the aforementioned parameters is depicted in Fig. 3.

$$T_d = \frac{60}{S \times M}. \quad (2)$$

For any radar simulation to be realistic, all the parameters listed in Table 1 as well as their the temporal relationship between them as it is depicted in Fig. 3 should be properly incorporated into the simulation tasks.

$$n_p = \left\lfloor \frac{T_d}{T_p} \right\rfloor. \quad (3)$$

3.2. LTE preliminaries

LTE is a world-wide wireless broadband standard, which enables ubiquitous cellular service, improved system capacity and coverage, better user experience through

Table 1

Typical operating parameters for radars.

Parameter	Typical value
Transmitter power	50 kW–2 MW
Pulse width	0.1–700 μ S
Pulse repetition interval (PRI)	0.15–3 ms
Duty cycle	0.1%–40%
Antenna diameter	1–8 m
Antenna rotation	6–60 rpm
Antenna gain	25–45 dBi
Azimuth beamwidth	0.6–3°
Elevation beamwidth	0.6–1.5°
Dwell time	3–80 ms
Polarization	Linear, circular

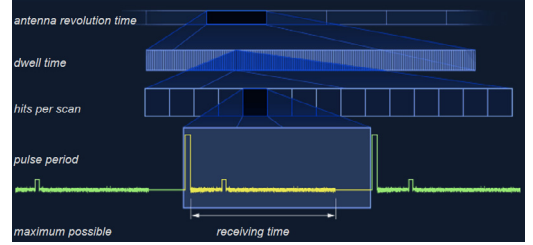


Fig. 3. Radar timing [19]: Each radar antenna rotation includes many dwell times during which radar pulses are emitted into the space.

higher data rates and reduced latency, seamless integration with the legacy communications systems, and reduced operational and deployment costs. LTE supports advanced multi-antenna schemes such as transmit diversity, spatial multiplexing, and beamforming. To attain superior Peak-to-Average Power Ratio (PAPR) characteristics on the UL, LTE leverages Single Carrier Frequency Division Multiple Access (SC-FDMA) [20] modulation in its UL direction. It can further operate in Frequency Division Duplex (FDD) and Time Division Duplex (TDD) modes.

Another characteristic of LTE is its capability to perform fast link adaptation based on the channel quality feedback from UEs to the serving eNB. In particular, Quadrature Phase Shift Keying (QPSK), 16 Quadrature Amplitude Modulation (QAM), and 64 QAM with coding rates varying from 0.07 to 0.93 are supported. LTE uses turbo coding for data transmissions.

Let $h_{s,f}$ denote the UL channel from an UE to its serving eNB at the time of transmission of symbol s on subcarrier frequency f . For simplicity, we assume a single transmit antenna at the UEs and a single receive antenna at the eNB. Extension to multiple transmit antennas at the UEs and eNBs is straightforward. It is well known [20] that UL SC-FDMA transmissions are demodulated by first applying single-tap frequency-domain equalizer $e_{s,f}$ given by

$$e_{s,f} = \frac{h_{s,f}}{|h_{s,f}|^2 + \sigma^2} \quad (4)$$

where σ^2 denotes noise-plus-interference power observed on subcarrier f . Following frequency-domain equalization, the single-carrier symbols are recovered by applying the inverse fast Fourier transform (IFFT) on the equalized frequency-domain samples. The resulting signal-to-

interference-to-noise-ratio (SINR) in symbol s is given by equation below [20].

$$SINR_{SC-FDMA,s} = \left(\frac{1}{N_f} \sum_{f=1}^{N_f} \frac{1}{SINR_{s,f}} \right)^{-1}. \quad (5)$$

Here, $SINR_{s,f} = \frac{|h_{s,f}|^2}{\sigma^2}$ is the pre-equalization SINR observed on subcarrier f , and N_f is the number of subcarriers allocated for the UL burst. Following the per-symbol SINR calculation, the burst error rate can be derived by applying exponential effective SNR mapping (EESM) [20] link-to system mapping across symbols assigned to the UL burst.

Moreover, based on the 3GPP-defined deployment scenarios [20], LTE can have several layouts related to the urban/suburban/rural macro cell, outdoor urban small cell, and indoor small cells, which are referred to as UMa/SMa/RMa, UMi, and indoor hotspot (InH) in the 3GPP documentation in that order. In this study, we look at UMa, UMi, and InH models for the cellular system. The propagation model within the cellular system is according to Eq. (6) [20].

$$\begin{aligned} P_L(f, d) = & [(16.9 \log(d) + 32.8) I(LoS) \\ & + (43.3 \log(d) + 11.5) I(NLoS)] \\ & + 20 \log(f) I(InH) \\ & + [(22 \log(d) + 28 + 20 \log(f)) I(LoS) \\ & + (36.7 \log(d) + 22.7 \\ & + 26 \log(f)) I(NLoS)] I(UMi) \\ & + \left\{ (22 \log(d) + 28 + 20 \log(f)) I(LoS) \right. \\ & + \left[161.04 - 7.1 \log(W) + 7.5 \log(h) \right. \\ & - \left. \left(24.37 - 3.7 \left(\frac{h}{h_{eNB}} \right)^2 \right) \log(h_{eNB}) \right. \\ & + (43.42 - 3.1 \log(h_{eNB})) \\ & \times (\log(d) - 3) + 20 \log(f) \\ & - 3.2 \log^2(11.75 h_{UE}) \\ & \left. \left. + 15.90 \right] I(NLoS) \right\} I(UMa). \end{aligned} \quad (6)$$

In this equation, $I(\cdot)$ is the indicator function which is 1 at the given argument and zero otherwise. For example, $I(LoS) = 1$ when we are at LoS and is zero if we are in NLoS, whereas $I(NLoS)$ is 1 at NLoS and is 0 when the UE is at LoS. On the other hand, $I(InH)$ is only 1 when we are in indoor small cells, whereas $I(UMa)$ is 1 when the scenario is urban macro (out door cells), and $I(UMi)$ is 1 when the scenario under investigation is an outdoor small cell. In the equation above, h_{UE} and h_{eNB} are UE and eNB antenna heights respectively. In 3GPP, these are usually taken as 1.5 m for the UE and 6, 10, 25 m for eNB in InH, UMi, and UMa layouts, in that order. The layout for the UMa scenario is the hexagonal grid and the traffic is full-buffer.

The simulation results show the layouts in Figs. 8, 10, and 11, respectively. We will provide with further explanations on the layouts in Section 4.3.

The eNB antenna pattern per sector for the UMa is given in Eq. (7) [7] where G_A and θ_A (G_E and θ_E) are respectively the antenna azimuth (elevation) pattern and azimuth (elevation) angle off the boresight [7]. In this equation, $-180^\circ \leq \theta_A \leq 180^\circ$, $-90^\circ \leq \theta_E \leq 90^\circ$, antenna azimuth and elevation downtilts are $\theta_{A,t} = 0^\circ$ and $\theta_{E,t} = 12^\circ$ respectively, $A_m = 20$ dB is the maximum attenuation, and $\theta_{3\text{ dB}}$ is the antenna 3 dB beamwidth. The normalized composite antenna pattern is expressed in Eq. (8).

$$G_i(\theta_i) = -\min \left\{ 12 \left(\frac{\theta_i - \theta_{i,t}}{\theta_{3\text{ dB}}} \right)^2, A_m \right\} \quad (7)$$

$$G = -\min \{ -(G_A(\theta_A) + G_E(\theta_E)), A_m \}. \quad (8)$$

For an LTE simulation to be representative, all the aforementioned parameters should be taken into account.

3.3. Propagation loss between radar and LTE system

The RF signals suffer from a propagation loss as they travel in the space. The intensity of the propagation loss depends on the distance traveled by the RF signals, the type of the environment, and seasonal effects. Different models for propagation are proposed; however, FCC and NTIA have long chosen FSPL and ITM as paramount models for the LoS and NLoS regions [4]. In fact, NTIA's Fast Track Evaluation [4] leveraged the FSPL and ITM for respectively the LoS and NLoS region propagation losses that the radar signals undergo before they reach the commercial system to which the interference effects were evaluated. To contrast the effect of radar into LTE systems with those of the Fast Track Evaluation [4], we use the same models, i.e. FSPL and ITM, for the radar-to-LTE propagation loss calculations.

FSPL can be expressed as Eq. (9) where f is the radar operating frequency in MHz, d is the distance in km at which the loss is of interest, $L_{\text{dB,FSPL}}$ in dB is the path loss attenuation, and r_{LoS} is the LoS region border in kilometers as Eq. (10) [4]. Here, where h_{radar} and h_{LTE} are the radar and LTE antenna heights in meters. Eq. (10) indicates that the LoS region border increases as the terminal heights increase. As an illustration, the LoS region from the Fast Track Evaluation is plotted in Fig. 4(a). We observe the LoS region where ITM (green, blue, yellow, red, and cyan curves) and FSPL (black curve) coincide. In the NLoS region, the ITM model predicts that propagation losses significantly rise compared to the LoS region; this reflects the diffraction loss. On the other hand, the NLoS region reflects the troposcatter loss as the distance further increases and observes a lower rise compared to the diffraction loss. The increase in LoS border by increasing the TX and RX antenna heights is depicted in Fig. 4(b).

$$L_{\text{dB,FSPL}} = 20 \log(f) + 20 \log(d) + 32.45 + p_L, \quad (9)$$

$$r \leq r_{\text{LoS}}$$

$$r_{\text{LoS}} = 4.1 \left(\sqrt{h_{\text{radar}}} + \sqrt{h_{\text{LTE}}} \right) \quad (10)$$

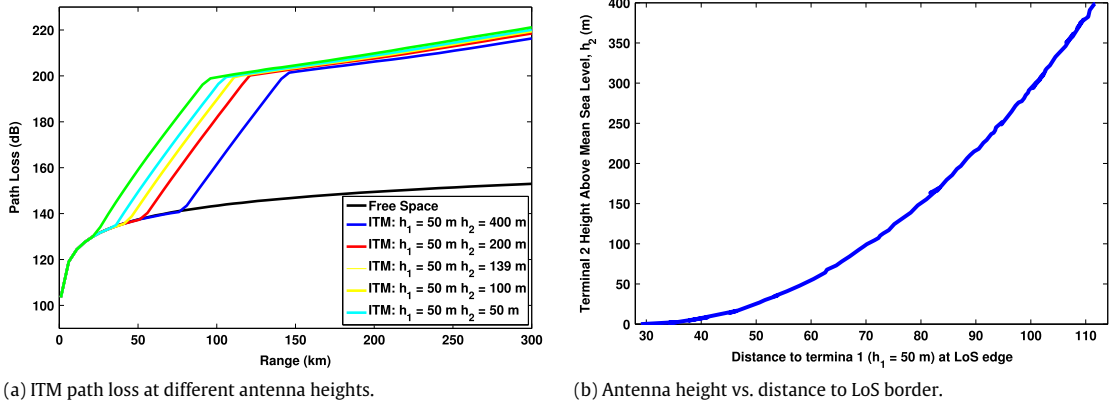


Fig. 4. LoS region border for the RX antenna height fixed at $h_1 = 50$ m when the other antenna height changes is illustrated. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

where p_L denotes the radar-to-eNB penetration loss to be applied if the eNB is deployed indoors. In the simulation results, penetration loss of 20 dB is assumed.

On the other hand, ITM accounts for environmental parameters such as ground conductivity in Siemens per meter (S/m), surface refractivity in parts per million, ground permittivity in Farad per meter (F/m), and average height change in meters. The ground attributes define the ground conductivity (0.005 for average ground, 0.001 for poor ground, 0.02 for good ground, 0.01 for fresh water, and 5.00 for sea water [9]) and permittivity (15 for average ground, 4 for poor ground, and 25 for good ground, fresh water, and sea water [9]). Radio climate also influences the ITM model and is divided into Continental Temperate, Continental Subtropical, Maritime Temperate (sea or land), Equatorial, Maritime Tropical, Desert are input to the model. The choice of the climate is based on the climate zones [9]. Besides, TX/RX antenna heights in meters and their polarization (vertical or horizontal), and antenna sitting criteria (random, careful, very careful) [9] are taken into the equation and air refractive gradient index, and climate to predict the amount of the loss using sophisticated statistical techniques [9].

As an illustration, the ITM predicted losses from the Fast Track Evaluation are depicted in Fig. 5. As we can observe, the propagation loss from the ITM and FSPL coincides in the LoS region and the amount of loss significantly rises with the traveled distance in the NLoS region which reflects the diffraction loss. Furthermore, the loss is significantly higher than the diffraction loss or the FSPL, but the amount of the loss increments is lower than the diffraction loss. This reflects the troposcatter loss in the NLoS region. The simulation code for the ITM model can be obtained from [9].

4. Simulation setup

In order to study the interference from radars into LTE systems, we simulate a radar system through modeling all the technical operating parameters in Section 3.1. The radar simulation includes the parameters from the NTIA's Fast Track Evaluation [4] as well as additional parameters for rotation and dwell time, which make the

radar simulation more realistic. In addition, we simulate an LTE system compliant to the 3GPP. The FSPL and ITM propagation path loss models are simulated.

4.1. Radar interference modeling

In this subsection, basic UL SINR models introduced in Section 3.2 are extended to include radar interference. Let $R_{s,f}$ denote frequency-domain radar receive signal observed at a UEs serving eNB. We note that in general

$$R_{s,f} = \Gamma_f T_{s,f} \sqrt{P_R A_{RX} A_{R,TX,s}} \quad (11)$$

where P_R denotes radar transmit power, Γ_f denotes the frequency-domain channel from the radar transmitter to the eNB receiver, A_{RX} denotes the receive antenna gain (computed with respect to the line-of-sight direction from the radar transmitter to the eNB receiver), $A_{R,TX,s}$ is the radar transmit antenna gain, and finally $T_{s,f}$ is the frequency-domain transmit radar signal in symbol s (normalized to unit average power). Note that the radar transmit antenna gain is time varying (function of symbol index s) due to the radar rotation. To the best of knowledge, frequency-domain and spatial characteristics of the radar-to-eNB channel are not well-understood and remain topic of ongoing research efforts. In this work, we have assumed that radar-to-eNB channel is frequency-flat and only consists of the propagation loss component. That is, $\Gamma_f = \Gamma$ and $|\Gamma|^2 = f(d)$ is the distance-dependent propagation loss between the radar transmitter and the serving eNB. Our goal is to extend the SINR expression in 3.2 with the effects of radar interference on the UL SC-FDMA receiver. To this end, we first define Signal-to-Radar Interference Ratio (SRIR) in symbol s as Eq. (17).

$$SRIR_s = \frac{P_{desired,s}}{P_{radar,s}}. \quad (12)$$

Here, $P_{desired,s}$ is the desired signal radiated from the UE and received at the eNB, and $P_{radar,s}$ is the radar signal received at the eNB. Given the SRIR definition above, we note that the final SINR expression can then be expressed as

$$SINR_{final,s} = \left(\frac{1}{SINR_{SC-FDMA,s}} + \frac{1}{SRIR_s} \right)^{-1}. \quad (13)$$

What remains is to compute the expressions for the desired and radar interference powers in the SRIR expression. Specifically, we note that the equalized desired signal power is given as

$$P_{desired,s} = \frac{P_D A_{RX} A_{D,TX}}{N_f} \left| \sum_{f=1}^{N_f} e_{s,f}^* h_{s,f} \right|^2 \quad (14)$$

where P_D is the UEs transmit power, A_{RX} is the eNB antenna gain, $A_{D,TX}$ is the UEs transmit antenna gain, and $e_{s,f}^*$ is the complex conjugate of the frequency response of the equalizer. The radar equalized receive power is similarly given as Eq. (15).

$$P_{radar,s} = \frac{P_R A_{RX} A_{R,TX}}{N_f} |\Gamma|^2 \left| \sum_{f=1}^N e_{s,f}^* T_{s,f} \right|^2. \quad (15)$$

The preceding expression gives the average post-equalized radar interference power in a given OFDMA symbol. We note that a more precise radar interference model can be modeled by considering radar interference in time-domain. Let index k denote the k th time-domain symbol transmitted during the given OFDMA symbol s . Then the radar interference observed during the k th time-domain symbol is given as

$$P_{radar,s,k} = P_R A_{RX} A_{R,TX} |\Gamma|^2 |IFFT[e_{s,f}^* T_{s,f}]_k|^2 \quad (16)$$

where $IFFT[\cdot]_k$ denotes the k th output of the IFFT transform. Also, the desired receive power, $P_{desired,s}$, is independent of k , so that the final SRIR expression during the k th time-domain symbol is given as

$$SRIR_{s,k} = \frac{P_{desired,s}}{P_{radar,s,k}}. \quad (17)$$

Finally,

$$SINR_{final,s,k} = \left(\frac{1}{SINR_{SC-FDMA,s}} + \frac{1}{SRIR_{s,k}} \right)^{-1}. \quad (18)$$

Given this final SINR expression for all s and k that comprise the UL burst, an EESM link-to system mapping [21] is used to derive burst error rate. In the event that a burst is declared in error, an UL HARQ process is initiated.

4.2. Radar simulation

The radar simulation contains the entire operating parameters in Section 3.1 with the values adopted from the NTIA's Fast Track Evaluation [4]. However, the parameters marked with an asterisk were not disclosed in the aforementioned report due to the sensitive nature of the radar operation and military classifications. For those parameters, we chose typical operating values for medium-to-large shipborne S-band radars [18]. For instance, we selected an angular speed of 30 rpm to make sure that every $T_s = \frac{60}{30} = 2$ s observes one complete rotation of the radar antenna. Horizontally scanning 360° with the speed 30 rpm, during the $T_s = 2$ s scan time of the radar $T_s \times f_p = \frac{T_s}{T_p} = 4000$ pulses are

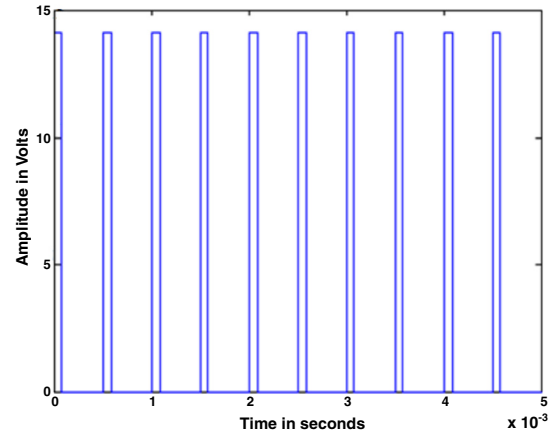


Fig. 5. Radar pulses during an antenna dwell time 4.5 ms over which the pulses of width 78 μ s with an PRI of 0.5 ms are radiated into the air from the antenna. Therefore, 9 pulses are transmitted during each dwell time.

emitted. The pulse power is 83 dBm are emitted which is roughly 200 kW. For other parameters, the azimuth beamwidth $\theta_H = 0.81^\circ$ necessitates $M = \frac{360}{\theta_H} = 445$ beam positions to cover the 360° search fence. Therefore, the antenna dwell time, the scan time divided by the number of beam positions to scan, becomes $T_d = \frac{T_s}{M} = \frac{2}{445} = 4.5$ ms. With the PRI $T_p = 0.5$ ms, the dwell time $T_d = 4.5$ ms covers 9 pulses to be transmitted by the radar. The radar pulses during a singular dwell time are depicted in Fig. 5. The pulsewidth is 78 μ s, and the PRI is 0.5 ms, which is the time between consecutive pulses.

Similarly to [4], the radar antenna is written in Eq. (19) and plotted in Fig. 6 as the normalized gain in terms of the angle θ of the antenna boresight. The first, second, and third expressions of Eq. (19) are the theoretical directivity pattern, a mask equation for pattern deviation from the theoretical value at the side-lobe (-14.4 dB main beam), and back-lobe, respectively. As we can see, the back-lobe is fixed at 50 dBm below the main lobe. It is noteworthy that we are simulating the rotations as well, and the rotations during the dwell time are accounted for. In fact, as we will discuss in Section 5, the interference is calculated per each transmission time interval (TTI) of LTE which means that we are simulating rotations every TTI even during the dwell time.

$$G(\theta) = \begin{cases} \frac{\pi}{2} \left(\frac{\cos\left(\frac{68.8\pi \sin(\theta)}{\theta_3 \text{ dB}}\right)}{\left(\frac{\pi}{2}\right)^2 - \left(\frac{68.8\pi \sin(\theta)}{\theta_3 \text{ dB}}\right)^2} \right) \\ -17.5 \ln\left(\frac{2.33|\theta|}{\theta_3 \text{ dB}}\right) \\ -50 \text{ dB}. \end{cases} \quad (19)$$

Radar is placed at 50, 100, 150, and 200 km away from the LTE. Then, the radar transmits pulses into the LTE network as in Fig. 7. The width of the radar radiation D depends on the distance from the radar d and the horizontal beamwidth θ_H in accordance with Eq. (20).

$$D = 2d \tan(\theta_H) = 0.03d. \quad (20)$$

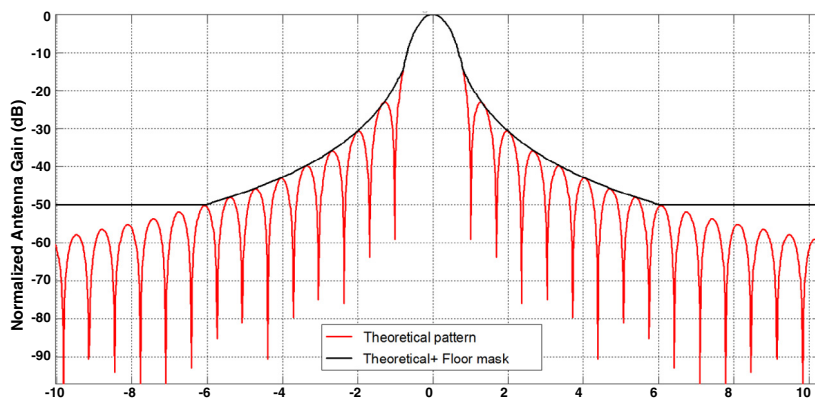


Fig. 6. Radar antenna pattern [4]: The black curve represents the pattern after the first side lobe is -24 dB. The back lobe is constant at -50 dB.

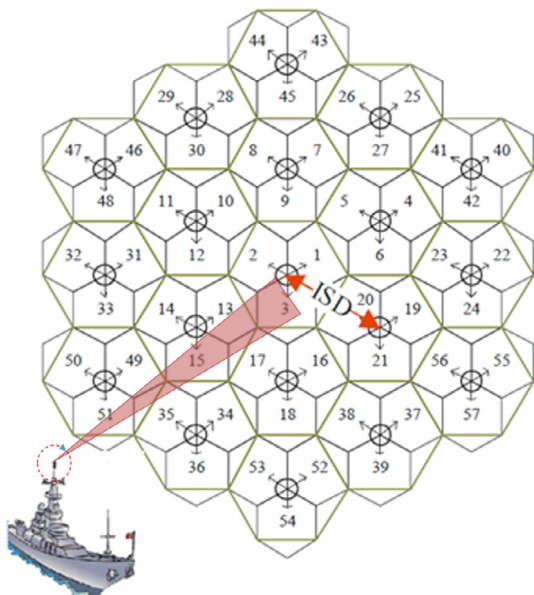


Fig. 7. Radar simulation scenario subsumes a radar next to the 3.5 GHz LTE.

Next, we describe the LTE simulation that we have used in this study.

4.3. LTE simulation

We leverage an LTE system level simulation which is 3GPP compliant [7]. It contains 3GPP-defined outdoor and indoor small cells as well as macro cell infrastructures. Furthermore, it utilizes a proportionally fair scheduler in both time and frequency domains, and includes a detailed UL air interface modeling, including UL multi input multi output (MIMO), RX diversity, non ideal link adaptation, and Hybrid Automatic Repeat Request (HARQ). For the results in this paper, a receive diversity combining Single Input Multi Output (SIMO) scheme with single transmit and two receive antennas (1×2 SIMO) is assumed. Furthermore, our LTE simulation has non-ideal link adaptation with HARQ and leverages an EESM link-to-system mapping [21]. In addition, we assume RF receiver

saturation at a threshold -50 dBm of wideband receive power and turbo decoder SINR saturation level of -30 dB.

The UL uses SC-FDMA [20], and deploys a full-buffer traffic model. Unless otherwise noted, radar signal is assumed to be co-channel with the LTE system. However, we also provide some results where radar carrier frequency is offset from the LTE systems carrier frequency.

First, we deployed the LTE simulator with a macro cell model as in Section 3.2, referred to as urban macro (UMa) [7], with an intersite distance 500 m. The macro cell layout simulations are hexagonal and are depicted in Fig. 8. As we see, there are 7 sites with 7 red circles representing the eNBs. The sites are sectorized with 120° sectors at 90° , 210° , 330° sectors. There are 3 cells per site which yields in 21 cells in the entire system. The eNBs antenna in the macro cell system are 25 m high, their gains are 17 dBi, and their downtilt is 12° . The UEs, depicted with the black addition symbols, are uniformly distributed around the eNBs and move with a speed of 3 km/h. The cellular system spans an area about $1600 \text{ m} \times 1600 \text{ m}$. The UE traffic that we use in the LTE simulation is full buffer, and there are 10 UEs in each cell which amounts to 210 UEs in the system, which are at least 25 m away from the eNBs. UEs have omnidirectional 0 dBi antenna. In the UMa simulation, 80% of the UEs are indoor, LTE bandwidth is 10 and 20 MHz, and it operates at 3.5 GHz in TDD mode with a UL:DL split 2:3. The eNB and UE transmit powers are 46 dBm and 23 dBm respectively and, considering the 17 dBi eNB antenna gain and 0 dBi UE antenna gain, the EIRP transmit power is maximally 63 and 23 dBm in that order. Moreover, the eNB and UE antenna heights are 25 m and 1.5 m, and their noise figures are 5 dB and 9 dB respectively.

Furthermore, the eNBs antenna pattern per sector which was given in Eq. (7) in Section 3.2 is depicted in Fig. 9.

Second, we did an LTE simulation with outdoor small cells, referred to as urban micro (UMi) [7]. One of the main distinctions of the UMi from the UMa is that its cells are not sectorized, are omnidirectional, shorter than the UMa eNBs, and have lower gains. The layout in our simulations is depicted in Fig. 10. Here, the blue squares represent the macro cell eNBs operating at 2 GHz, so they do not interfere with the small cells and the radar which operate in the 3.5 GHz band. The red circles show outdoor small

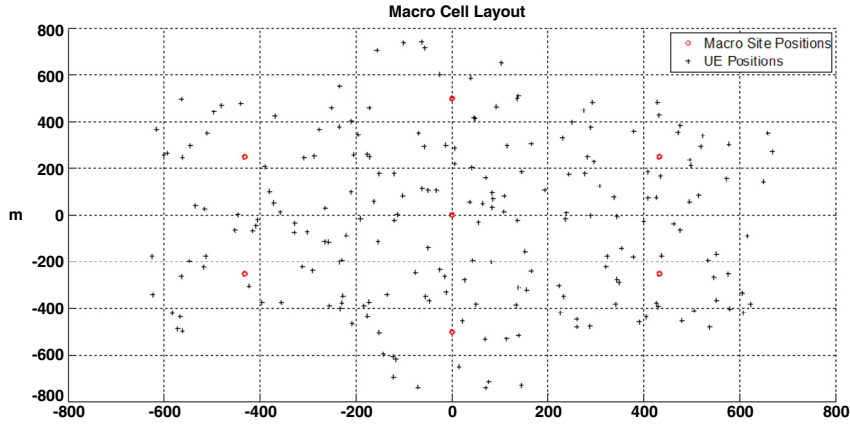


Fig. 8. LTE macro cell (UMa) hexagonal layout simulation of a 7 site, 21 cell system where each eNB is represented as a small red circle in the $1600 \text{ m} \times 1600 \text{ m}$ grid. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

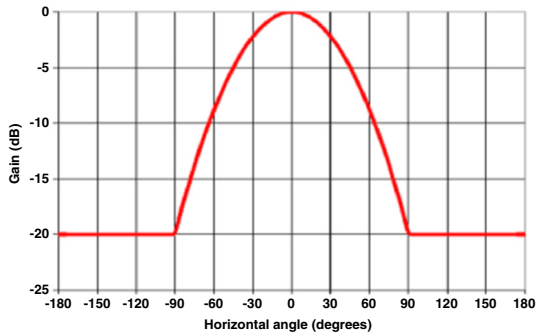


Fig. 9. Composite antenna pattern for the macro cell LTE eNBs for 3-sector cells from Eq. (3) [7].

cell eNBs, and their UEs are shown as the black addition symbols clustered about the eNBs. The layout spans an area as large as $1600 \text{ m} \times 1600 \text{ m}$. The operation is at 3.5 GHz in TDD mode with 2:3 UL:DL split. Amongst the UEs, 20% are indoor.

There are 84 small cells in the UMi simulation; the eNB gain is 5 dBi and it is 10 m high as opposed to the 17 dBi and 25 m antennae in the UMa model. Also, the minimum distance between the UEs and eNBs is 5 m as opposed to the 25 m UE-eNB distance in the UMa case. There are 30 UEs per small cell, which creates $30 \times 84 = 2520$ UEs in the UMi simulation. The UEs move with uniform direction at 3 km/h and are clustered around the eNBs. The LTE bandwidth in our simulation is 10 and 20 MHz. The modeled traffic is the same as the UMa, i.e. full buffer best effort. The eNB and UE transmit powers are 30 dBm and 23 dBm respectively and, considering the 5 dBi eNB antenna gain and 0 dBi UE antenna gain, the EIRP transmit power is maximally 35 dBm and 23 dBm in that order.

Third, we did an LTE simulation with indoor small cells, referred to as Indoor Hotspot (InH) [7]. One of the main distinctions of the InH from the UMa and UMi is that its layout spans over a hallway as in Fig. 11. Here, the red circles show indoor small cell eNBs; we see a $50 \text{ m} \times 120 \text{ m}$ corridor with two eNBs at 30 and 90 m alongside the hallway. The UEs, shown by the plus symbols in Fig. 11, are scattered around the eNBs, and there are 20 UEs in

our simulation which move at 3 km/h. We mentioned that there are 10 UEs per cell, which in light of the fact that LTE indoor has 2 cells (2 eNBs)—Fig. 11, it becomes the 20 UEs for the simulation. The eNBs are omnidirectional, are 6 m high which are shorter than the UMi eNBs, and have 5 dBi gains. The operation is at 3.5 GHz in TDD mode with 2:3 UL:DL split.

The LTE bandwidth in our simulation is 10 and 20 MHz. The modeled traffic is the same as the UMa and UMi, i.e. full buffer best effort (FBBE). The eNB and UE transmit powers are 30 dBm and 23 dBm respectively and, considering the 5 dBi eNB antenna gain and 0 dBi UE antenna gain, the EIRP transmit power is maximally 35 and 23 dBm in that order.

4.4. Propagation model between radar and LTE

Similarly to the Fast Track Evaluation [4], we simulate the ITM in its area prediction mode (APM) [9] in order to model the NLoS region propagation loss. We use APM, as opposed to the point-to-point mode (P2PM), to provide with general results of propagation loss based on the terrain irregularity parameter [9]; the P2PM is based on the precise elevation profile on any propagation path which makes it a great choice for cell planning. The goal of this article to provide with a view on radar-LTE shared-spectrum operation without worrying about precise elevation values and instead focusing on terrain elevation average changes. The ITM provides with the diffraction and troposcatter losses introduced onto the radar signals in the NLoS region which starts after the LoS border. Here, where the radar antenna height $h_{\text{radar}} = 50 \text{ m}$ (large shipborne radar) and the LTE antenna heights for InH, UMi, and UMa $h_{\text{LTE}} = 6, 10, 25 \text{ m}$ cause the LoS to be effective until 39.0 km for InH, 42.0 km for UMi, and 49.5 km for UMa. This means that in all distances away from the radar until the 49.5 km for macro cells and 42.0 km for outdoor small cells, and 39.0 km for indoor small cells, the propagation loss comes from the FSPL model.

Similarly to [4], the ITM for the loss in the NLoS region in its area prediction mode (APM) [9]. The terrain roughness is set to 10 m, and the antenna heights for LTE macro, outdoor LTE small cell, and indoor LTE small cell are

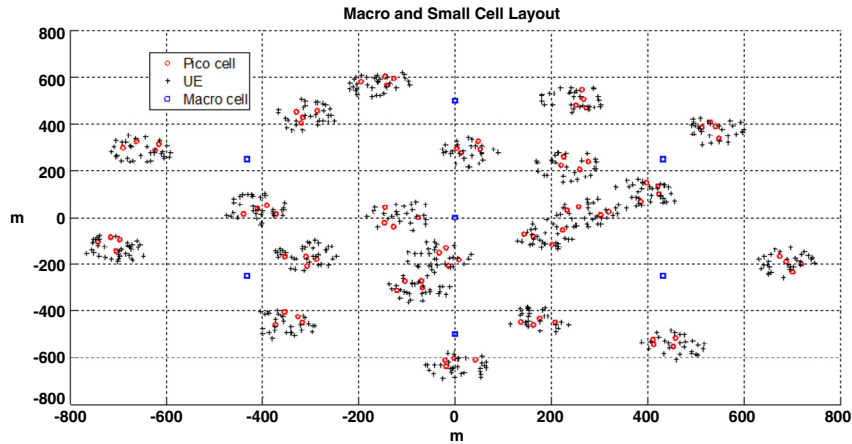


Fig. 10. LTE micro cell (UMi) layout with 84 cells where the eNBs are shown as red clustered circles [7] and the UEs are the gray plus signs uniformly distributed. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

set to 25, 10, and 6 m, respectively. The radar antenna height is 50 m, ground dielectric constant is 15, ground conductivity is 0.005 S/m, refractivity is 301 N-units, the climate is continental temperate, and single message mode is used in compliance with [4]. Furthermore, the confidence parameter in our model is chosen as 50%. The plots for the FSPL and ITM losses are depicted in Fig. 12. Here, the cyan curve indicates FSPL, red curve is the ITM loss for the macro cell case (UMa), the green curve is the ITM loss for the outdoor small cell case (UMi), and the blue curve is the ITM loss for the indoor small cell case (InH). The vertical axis is the pathloss in dB and the horizontal axis is the distance. As we see from Fig. 12, initially pathloss increases using the FSPL model, then it escalates dramatically and almost linearly in the NLoS region which is the diffraction loss [9], and ultimately the loss grows less rapidly which is the troposcatter loss. It is worth mentioning that the FSPL and ITM models predict very close propagation losses in the LoS region whilst the introduced loss sharply elevates in the NLoS region, where ITM model is utilized.

Before delving into the simulation results in Section 5, a high-level diagram of the LTE system level simulator that we leveraged for the simulation of the LTE network is presented in Fig. 13, which summarizes the descriptions of the LTE simulation as we discussed in Section 4.3. The first step is to place base stations on the cellular grid according to the UMa, UMi, or InH system layouts. This is followed by placing UEs randomly and uniformly in the entire cellular system area. Next, the propagation channel between the UEs and the eNBs is calculated according to the UMa, UMi, and InH channel models. Once the channels are computed, we assign the UEs to the serving eNBs according to the max-SNR metric. Next, we place the simulated radar (Section 3.1) at various distances from the geometric center of the cellular system. In each subframe, UL transmissions from the UEs to the serving eNBs are scheduled. In each symbol of the subframe, we calculate the radar beam position (as the radar rotates), and based on the relative angles between sector orientation and the current radar beam position compute SINR per symbol for all scheduled UL bursts. Then, we use the link-to-system

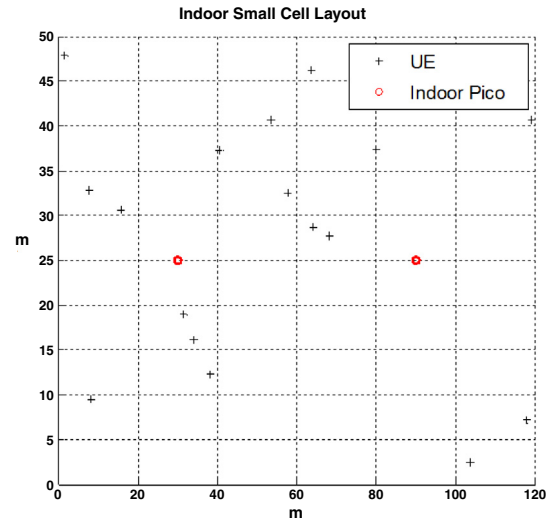


Fig. 11. The layout for indoor small cells (InH) in our simulation, which shows two eNBs as the red circles at positions 30 and 90 m along the 50 m $\times$ 120 m corridor [7].

level mapping [21] to obtain the frame error rate (FER) for the scheduled bursts, with bursts that are in error to be retransmitted. At the completion of the subframe loop, the throughput for the LTE is calculated and the simulation ends.

Next, we present the simulations results for the radar-to-LTE interference in the UL directions for macro cells, outdoor small cells, and indoor small cells.

5. Simulation results

We set the simulation time to 5 s, during which the impact of the radiations from the radar with parameters in Section 4.2 into the eNBs of an LTE cellular system with parameters in Section 4.3 is investigated. We consider both cases where the radar with parameters in Section 4.2 is co-channel or out-of-band with respect to the LTE system and rotates 360° in azimuth covering 445 beam positions,

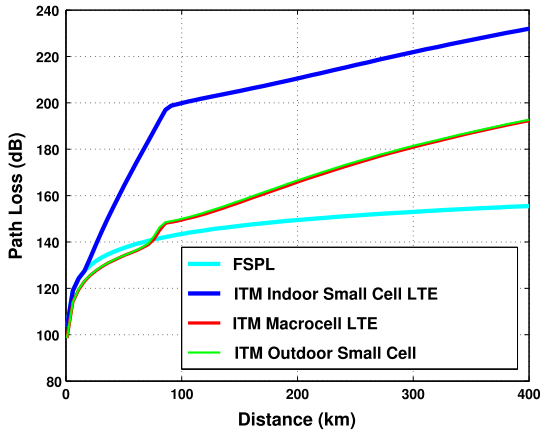


Fig. 12. LoS FSPL in green and NLoS ITM loss in blue and red for macro and small cells respectively represent the radar signal degradation in dB vs. distance in km. The first nonlinear part of the curve starts with the FSPL which occurs in LoS region, and then the path loss deviates from the FSPL into scattering loss and ultimately into the troposcatter loss region. For distances more than 70 km between the radar and LTE wave propagation experiences the troposcatter loss. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

where the radar sojourns for 4.5 ms and transmits 9 pulses each 78 μ s wide and 83 dBm through its 45 dBi antenna to the eNBs. The eNBs covered by the radiation will suffer from the radar pulses which are amplified by the TX and antenna gain and undermined by the propagation losses explained in Section 4.4. Next, we present the simulation results for the macro cells, i.e. the UMa model, in Section 5.1.

5.1. Radar-macro cell LTE UL interference

With regard to the interference, the main performance metric for LTE systems is the mean throughput [4,6,8]. We assume that the radar is cochannel with the UMa LTE system which operates with a 10 MHz bandwidth under the FDD mode. The normalized mean UL throughput as a function of the distance of the radar from the LTE system is depicted in Fig. 14 as a purple curve. The ordinate in Fig. 14 is the UL mean throughput relative to the baseline, i.e. no radar is present, shown as the value 1 on the ordinate. The abscissa shows that we placed the radar at 50, 100, 150, and 200 km from the LTE system. As we see from the purple curve, slight throughput losses incur when the radar is 50, 100, 150, and 200 km away from the radar vis-à-vis the baseline which is the value 1. In fact, the LTE system as close as 100 km away from the radar undergoes less than 10% throughput loss with respect to the baseline, and the loss is less than 30% when the radar is only 50 km away from the LTE. As we can see, the LTE macro cells are very robust to radar interference even when LTE is experiencing extreme signal-to-interference-plus-noise ratio (SINR) conditions. Moreover, for this 10 MHz LTE, we plot the SINR of an LTE macro eNB vs. LTE symbol and subcarrier indices in Fig. 15. As we can observe, SINR drops due to radar pulses affecting LTE symbols in UL. Interestingly, even when the radar is present, the SINR

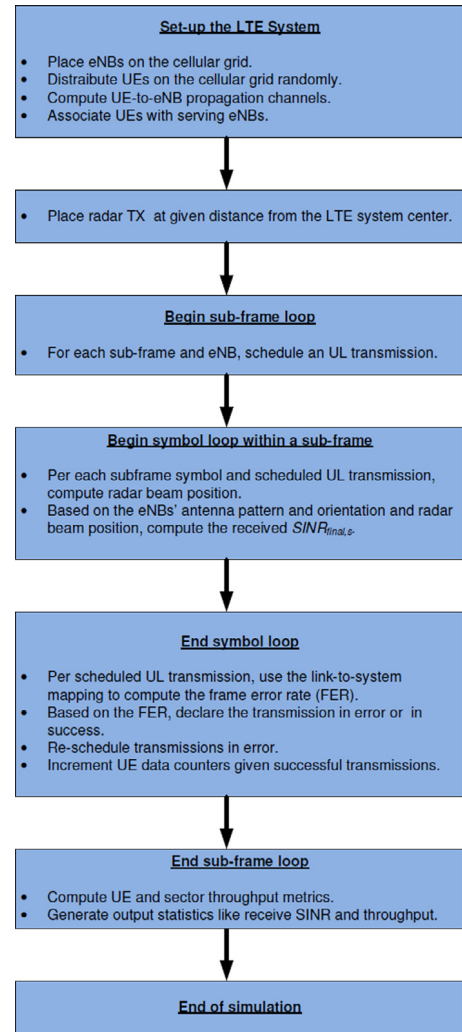


Fig. 13. LTE system level simulation steps.

recovers back to its normal baseline situation until the next pulse hits the same region. For instance, the radar hits symbol 1 and since the radar pulses (78 μ s) are larger than the LTE symbols (71.4 μ s), symbol 2 is also affected. Then, the next pulse hits symbol 8, which spans over to symbol 9 too for the same reason. So, most of the radar energy is concentrated on symbol 1 and symbol 8, with the remaining pulse energy present on symbols 2, 9 and 14. This is promising as only certain symbols within an LTE sub-frame are affected by the radar. Also, there are 7 symbols between symbol 1 (start of the first radar hit) and symbol 8 (start of the second radar hit), corresponding to $7 \times 71.4 = 0.5$ ms, which is the radar PRI as expected. Because the radar signal is assumed to be centered in the LTE band, most pulse energy is around subcarrier 300, i.e. in the middle of the LTE band.

Furthermore, we illustrate the mean UL throughput for a 20 MHz TDD UMa LTE as the red curve in Fig. 15. As we can observe, until 50 km away from the radar, LTE suffers from less than 20% throughput loss compared to the baseline. The amount of throughput loss at 100, 150, and 200s km

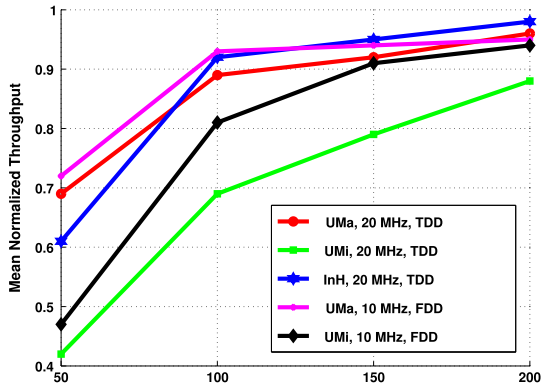


Fig. 14. UL throughput for 10 MHz FDD and 20 MHz TDD LTE and cochannel radar at various distances. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

away from the radar is small. The UL throughput CDF for the 20 MHz TDD LTE UMa when the cochannel radar is at 50, 100, 150, and 200 km away from the LTE system is also depicted in Fig. 18(a). As we can observe, the closer the radar to the LTE system, the curves shift to the left further which indicates a throughput loss increase.

We also look at the out-of-band (OOB) interference from the radar into the 10 MHz FDD macro cell LTE system and its effect on the UL throughput of the system as the purple curve in Fig. 17. Here, the radar, lingering 50 km away from the LTE system, operates at a frequency 5 MHz, 10 MHz and 20 MHz offset from the LTE operating frequency. The throughput with respect to the baseline, i.e. no radar in the system, shown as the value 1 on the ordinate, drastically alleviates as the offset grows to 5 and 10 MHz. Moreover, we plot the UE UL throughput for an out-of-band radar 50 km away from the 20 MHz TDD LTE system as the red curve in Fig. 17. As we can see, the higher the offset, the lower the throughput loss compared to the baseline, where the radar is absent, represented by the value 1 on the ordinate. Zero MHz offset is equivalent to the cochannel situation in Fig. 14 at 50 km. The CDF for the OOB radar 50 km away from the 20 MHz TDD UMa

LTE system is illustrated in Fig. 16(a), which expectedly shows that 5 MHz offset yields in a higher loss as opposed to larger offsets. And, we can see that, for the higher offsets, the curves shift to the right side of the coordinate system further; this indicates an increase in the throughput of the system.

5.2. Radar-outdoor small cell LTE UL interference

Similarly to the macro cells, we look at the mean UE throughputs when the radar with parameters in Section 4.2 is 50, 100, 150, and 200 km away from the outdoor small cell LTE with parameters in Section 4.3. The normalized mean UE throughput relative to the baseline for a 10 MHz FDD LTE is illustrated as the black curve in Fig. 14. As we see, the throughput loss for the LTE small cells is not dramatic until 100 km away from the radar. In fact, for an LTE outdoor small cell as close as 150 km away from a cochannel radar the throughput loss is less than 10%, whereas at 100 km the throughput loss becomes slightly less than 20%. At 50 km way the throughput loss is approximately 55% in contrast to the baseline, represented by the value 1 on the ordinate in Fig. 14. We believe that the more decaying trend that we observe for the outdoor small cell LTE vs. the macro cell, i.e. black curve vs. the purple curve in Fig. 14 is because small cell LTE eNBs are omnidirectional and are not downtilted; so they are more exposed to the radar radiations and cause a more dramatic loss as the radar gets closer to the LTE system. As anticipated, the further away the radar, the higher the mean throughput as interference becomes less pronounced due to the propagation losses caused by the ITM in the NLoS region. Similarly, the UL throughput for a 20 MHz TDD outdoor small cell LTE when the cochannel radar is 50, 100, 150, and 200 km away from the LTE system is shown as the green curve in Fig. 14. As we can see, the throughput loss is rather significant at 50 km away. To further distinguish the UL throughput, we plot the UE throughput CDFs in Fig. 18(a), which shows that at 50 km the throughput loss is more dramatic than the other cases.

Next, we look at the OOB results for a radar at 50 km away from the 20 MHz TDD LTE outdoor small cells as

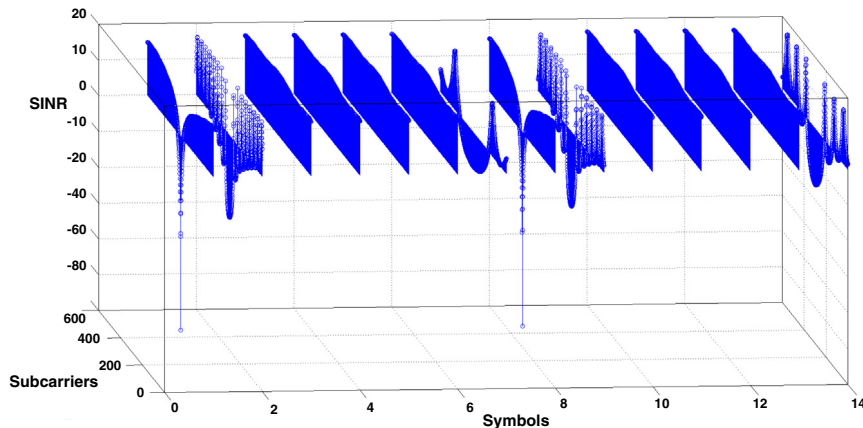


Fig. 15. SINR for the 10 MHz FDD LTE system in the UL direction when a cochannel radar at the center of the LTE band affects the eNBs. SINR recovers between the radar pulses and drops during the time the eNB is hit by a radar pulse. This is promising as only certain symbols are affected in an LTE subframe. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

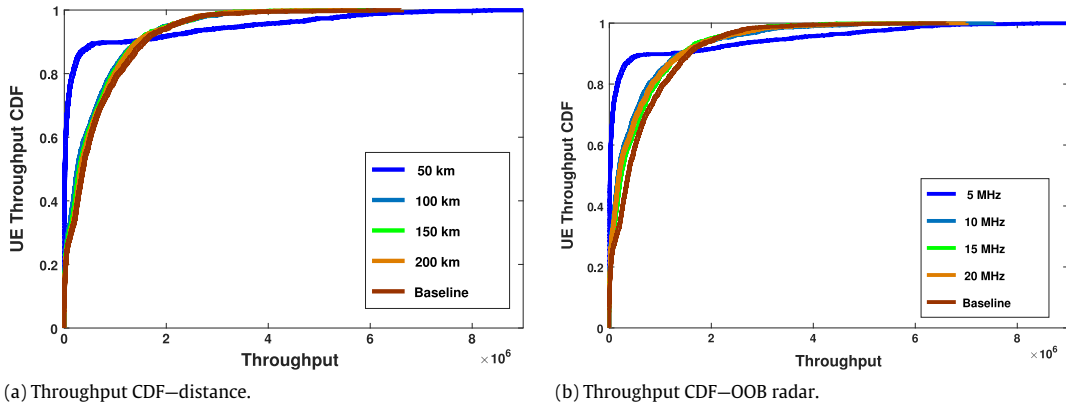


Fig. 16. UL throughput CDF for a 20 MHz TDD UMa LTE.

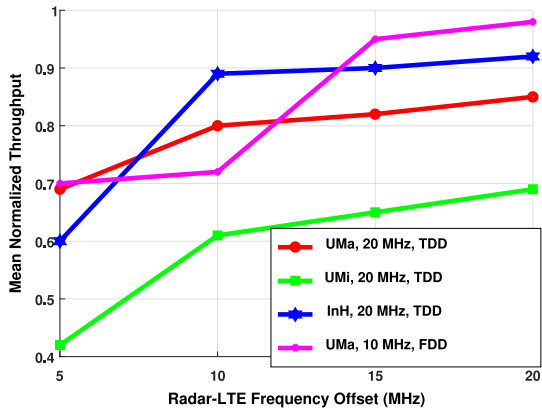


Fig. 17. UL throughput for 10 MHz FDD and 20 MHz TDD LTE and OOB radar at various frequency offsets. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

the green curve in Fig. 17, where the radar is 5, 10, 15, and 20 MHz offset from the LTE band center. As we can see, the throughput loss is more dramatic at 5 MHz offset; however, for larger offsets the throughput loss becomes less pronounced. The UL throughput CDFs of the 20 MHz TDD LTE with the OOB radar are shown in Fig. 18(b), which expectedly depicts that a 5 MHz offset accompanies the highest throughput loss in the system.

5.3. Radar-indoor small cell LTE UL interference

Here, we look at the mean UL throughput for the LTE indoor small cells (InH) when the radar with parameters in Section 4.2 is 50, 100, 150, and 200 km away from the LTE network with parameters in Section 4.3. The normalized mean throughput relative to the baseline, represented by the value 1 on the ordinate, is illustrated as the blue curve in Fig. 14 for the 20 MHz TDD LTE. As we observe, the throughput loss for the LTE small cells is not dramatic until 100 km away from the radar. In fact, for an LTE indoor small cell as close as 150 km away from a cochannel radar the throughput loss is less than 20%, whereas at 100 km the throughput loss becomes slightly less than 20%. At 50 km way the throughput loss is less than 40% in contrast to the

baseline. As it is anticipated, the further away the radar, the higher the mean UE throughput as interference becomes less pronounced due to the diffraction loss caused by the ITM in the NLoS region. Furthermore, the UL throughput CDF is depicted in Fig. 19(a). As we can observe, the CDF curves are close to one another for the baseline, 100, 150, and 200 km distance between the radar and LTE system. However, when the radar gets to 50 km vicinity of the LTE system, the CDF curve is significantly shifted to the left of the other curves. This shows the fact that throughput has dropped. Next, we look at the mean UL throughput for the indoor small cell 20 MHz TDD LTE system when the OOB radar is 50 km away from the LTE system, the blue curve in Fig. 17. A 0 offset represents a cochannel radar which is the same situation as the blue curve in Fig. 14 when the radar is 50 km away from the LTE system. However, as the radar operating frequency offsets from the LTE band center, the throughput loss decays so that at 20 MHz offset, there is less than 10% loss in the throughput. On the other hand, even, a 5 MHz offset, drastically increases the LTE throughput. Comparing this with the OOB results for UMi and UMa represented respectively as green and red curve in Fig. 17, we see that InH is more robust; this is expected due to the fact that indoor situations offer more loss to the received signal strength from the radar. And, the UL throughput CDF of the InH LTE for the OOB radar is depicted in Fig. 19(b), which shows the dramatic shift of the CDF to the left of the abscissa due to higher throughput loss at the 5 MHz offset.

6. Conclusion

In this paper, we studied the interference from a radar system into LTE cellular system. The radar is based on the operating parameters from the NTIA's Fast Track Evaluation [4]. The radar simulation includes rotation of the radar, antenna pattern, dwell time, pulse repetition interval, azimuth/elevation beamwidths, and transmit power for a practical radar.

Propagation model between the radar and LTE system was simulated based on the irregular terrain model's area prediction mode. Furthermore, a 3GPP compliant LTE system-level simulator was leveraged to create macro cell, outdoor small cell, and indoor small cell LTE systems by

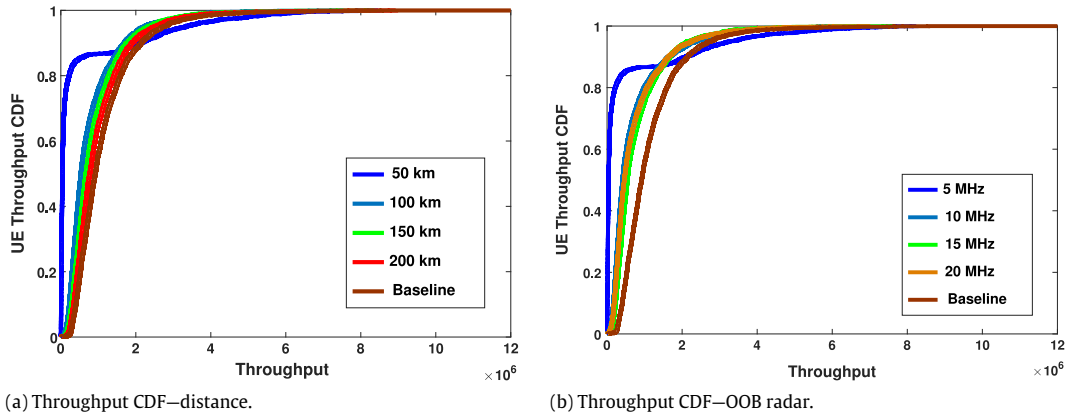


Fig. 18. UL Throughput CDF for a 20 MHz TDD UMi LTE.

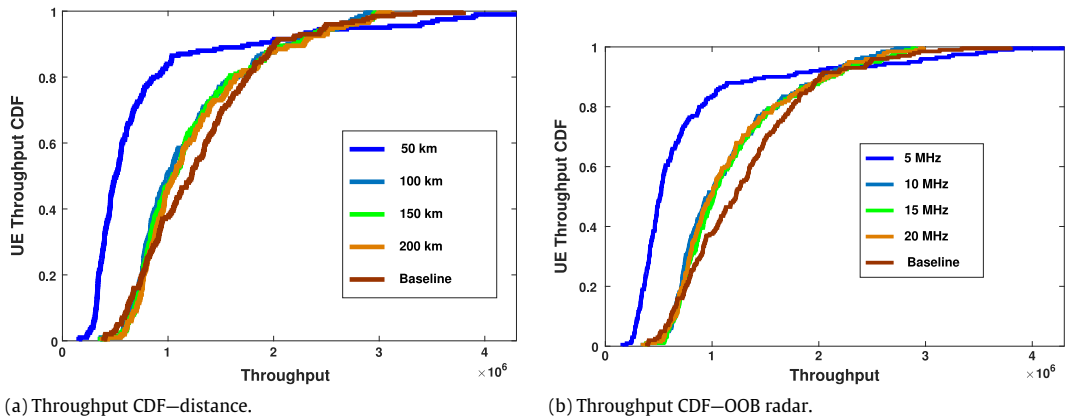


Fig. 19. UL Throughput CDF for a 20 MHz TDD InH LTE.

simulating all the details of the LTE protocol from the 3GPP standard. The propagation model within the LTE system was based on 3GPP stochastic models. The interference effects from the radar into the LTE eNBs was investigated at various distances between the radar and LTE network.

Furthermore, we look at the out-of-band interference from the radar into the LTE systems. Our simulations show that when the radar offset with respect to the LTE operating frequency is small, the amount of throughput loss for the LTE is very close to the cochannel radar. On the other hand, when the amount of offset approaches the LTE bandwidth, the throughput loss becomes negligible and very close to the baseline where no radar is present. Our simulation results imply that LTE is robust to the ship-borne radar interference deployed in 3.5 GHz even when LTE is experiencing extreme signal-to-interference-plus-noise ratio conditions. However, mean throughput losses were observed with radar situated 50 km or less from the system center.

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